

# Interorganizational Teams as Boundary Spanners Between Supplier and Customer Companies

Ruth Maria Stock

University of Stuttgart-Hohenheim

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*Extant research has largely ignored the phenomenon of interorganizational teams, which consist of members from both supplier and customer companies. This study examines the degree to which team interorganizationality influences team performance in a business-to-business context. On the basis of resource-dependence theory and boundary theory, the author argues that team interorganizationality positively influences team effectiveness, particularly when uncertainty is high. The hypotheses testing is based on multiple informant data collected from members and leaders of 225 teams in various industries. The results show the positive influence of team interorganizationality on team effectiveness. In addition, uncertainty-related moderator variables (company-related, market-related, and technological uncertainty) strengthen the link between team interorganizationality and team effectiveness.*

**Keywords:** *interorganizational teams; selling teams; new product development teams; marketing teams*

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Companies in business-to-business (B2B) markets continually face powerful and more demanding customers (e.g., Stock and Hoyer 2005). In many industries, business customers have reduced the number of suppliers and intensified their relationships with a chosen few (e.g., Cannon and Perreault 1999). In turn, they often demand special value-adding activities from their suppliers, such as joint product development, advanced personal interaction, or consulting services (Cardozo, Shipp, and Roering 1992; Homburg and Stock 2004).

These changing customer demands raise an organizational design problem for many suppliers. As Homburg, Workman, and Jensen (2002) stated, "Many suppliers rethink how . . . they design their internal organization in order to be responsive to . . . key customers" (p. 38). In this context, many supplier companies are shifting toward team-based structures to manage the boundaries with their customers (e.g., Perry, Pearce, and Sims 1999). Teams may be an effective response to the growing expertise demands of boundary-spanning employees in B2B relationships. "As products and services become more technologically complex, and as buyers come to expect increasing levels of service, team selling is becoming more prevalent" (Moon and Armstrong 1994:17).

Marketing research related to teams at the boundary between supplier and customer companies pertains mainly to three fields: sales management literature (e.g., Dixon, Gassenheimer, and Barr 1999), services marketing literature (e.g., de Jong, de Ruyter, and Lemmink 2003, 2004), and literature on new product development (e.g., Sethi 2000a, 2000b). Most sales management research focuses on selling teams, whereas services marketing research tends to consider teams of customer-contact employees who are jointly involved in the delivery of customer service. Product development literature focuses on new product teams. In a B2B setting, however, suppliers use interorganizational teams (i.e., containing members from both the supplier and the customer companies), such as joint new-product-development teams with important customers and joint customer service teams (e.g., a management consulting firm and a customer company).<sup>1</sup>

This article aims to provide a theoretical understanding of the phenomenon of interorganizational teams and the performance implications of team interorganizationality. In this study, team interorganizationality was conceptualized as the degree of interorganizationality rather

than as a simple yes-or-no variable. Specifically, two facets of team interorganizationality were considered: team composition and power distribution within a team. Given these two facets, a multiplicative index to measure team interorganizationality was built. Thus, team interorganizationality is highest when each organization is represented by the same number of team members and power is equally distributed between those organizations. The lowest level of interorganizationality occurs when a team consists of members of a single organization or power is totally concentrated in the hands of team members from only one organization.

The focal dependent construct of this study was team effectiveness, conceptualized from the supplier's perspective. However, it also addressed possible moderating effects on the link between team interorganizationality and team effectiveness, on the basis of the proposition that the suggested effect of team interorganizationality on team performance depends on certain contingency factors. Thus, two research questions were addressed:

1. Does the degree of team interorganizationality affect team effectiveness?
2. Which factors moderate the link between team interorganizationality and team effectiveness?

The analysis was based on data involving judgments by (multiple) members of marketing teams and their team leaders, which reduced the risk of a common method bias. To enhance the generalizability of the findings, data were collected from multiple companies in different manufacturing and services industries in a B2B context. Furthermore, this study included three types of teams: selling teams, customer service teams, and new-product-development teams.

## THEORETICAL BACKGROUND

### Resource-Dependence Theory

Resource-dependence theory views interfirm governance as a strategic response to conditions of uncertainty and dependence. Its key underlying assumption is that organizations are not internally self-sufficient with respect to their critical resources (Pfeffer 1982; Pfeffer and Salancik 1978), in that "they require resources from the environment and, thus, become interdependent with those elements of the environment with which they transact" (Pfeffer 1982:192-193). This lack of self-sufficiency creates potential dependencies on the parties from which the focal resources are obtained (Pfeffer 1982). This in turn introduces uncertainty, since the resource flows are not subject to the firms' control and may be difficult to predict.

Another main premise of resource-dependence theory is that organizations establish formal or semiformal links

with other organizations (specifically, owners of critical resources) to reduce environmental uncertainty and manage their dependence (Pfeffer 1982:200). These links with other organizations have been referred to as "bridging strategies" (Pfeffer 1982), whose main purpose is to gain control over the other organizations' resources.

In a B2B context, customer companies own resources that are relevant for supplier companies' survival (i.e., information demand). Thus, resource-dependence theory provides a fundamental theoretical explanation as to why the use of interorganizational teams (at the boundary with customer companies) may increase organizational effectiveness. According to the logic of resource-dependence theory, establishing such teams constitutes a bridging strategy. Interpersonal interactions within interorganizational teams can stabilize the relationship between a supplier and its customers through exchanges of information and reductions in uncertainty (Ancona and Caldwell 1990).

A further implication of resource-dependence theory relates to the identification of conditions in which establishing interorganizational teams should be particularly beneficial. For instance, resource-dependence theory suggests that a supplier company's dependence on its customers is particularly high in the case of high competitive intensity, high market-related uncertainty, and a low influence of the supplier on the resources held by the customer companies (Pfeffer 1982; Pfeffer and Salancik 1978).

### Boundary Theory

Using an open-systems perspective, boundary theory assumes that an organization depends on its environment for critical resource inputs, as well as for the disposal of its outputs (Aldrich and Herker 1977). From this perspective, a central challenge for organizations is to manage their boundaries with other organizations. An organizational boundary represents the domain of effort in which an organization interacts with its environment to survive (Aldrich and Herker 1977).

A second assumption of boundary theory is that an organization must protect itself from disruptive forces in the environment that would intrude on it. These disruptive forces depend on the level of environmental uncertainty (Cross, Yan, and Louis 2000). According to this theory, organizations engage into boundary-spanning activities, such as negotiation, contracting, and cooperation building, to manage their interactions across boundaries (Cross et al. 2000).

With respect to boundary-spanning activities, boundary theory emphasizes the crucial role of individual employees in establishing and maintaining interorganizational relationships. Theorists therefore have introduced the terms *boundary spanner* and *boundary-spanning employee* (Aldrich and Herker 1977) to refer to those employees who play multiple roles at the interface of an organization

and its environment. These roles include maintaining the image of the organization, functioning as information processors and filters (Aldrich and Herker 1977), influencing the organization's clients (Thompson 1967), and representing the perceptions, expectations, and ideas of each side to the other (Friedman and Podolny 1992).

Also according to boundary theory, the importance of boundary-spanning activities increases with greater uncertainty (Adams 1976). Whereas early research related to boundary theory used a more general notion of uncertainty, more recent research distinguishes among three types (Abernathy and Clark 1985): company-related, market-related, and technological.

## FRAMEWORK AND HYPOTHESES

### Overview of Framework

The unit of analysis for this study was a customer-related team in a B2B setting. As mentioned previously, such teams may include selling teams, customer service teams, or new-product-development teams. Team interorganizationality was the key construct of this study, whose conceptualization was based on work by Berné (1963). Important such boundaries include those between team leaders and team members and those between team members from different functional groups of the company, as well as between employees from different companies.

The prevalence of boundaries within a team can be described by two dimensions: team composition and power distribution (Berné 1963). The prevalence of boundaries is strongest when a team is composed equally of different subgroups and power is distributed evenly across them. In terms of team composition, interorganizationality is highest when each organization is represented by the same number of team members and lowest when the team consists solely of members of a single organization. Similarly, interorganizationality with respect to power distribution within a team is highest when power is distributed equally among the involved organizations and lowest if power is totally concentrated in the hands of members from one organization.

The dependent variable was team effectiveness, defined as the extent to which a team achieves its goals (Hackman 1987). This research conceptualized effectiveness from the supplier organization's perspective rather than the customer's perspective. The central proposition of the basic conceptual framework shown in Figure 1 was that team interorganizationality has a positive influence on team effectiveness. This proposition is rooted in both resource-dependence theory and boundary theory.

In addition to the main effect of team interorganizationality on team effectiveness, the framework incorporated selected moderator variables. In certain circumstances,

team interorganizationality may be more beneficial for a supplier company than it would be in others, so company-related, market-related, and technological uncertainty may alter the effect of interorganizationality.

The framework also included several control variables that presumably affect the dependent variable (i.e., team effectiveness) and have been identified as important drivers of team performance: team member characteristics, the characteristics of team design, and environmental factors. To represent team member characteristics, member expertise (e.g., Ancona and Caldwell 1992) was investigated. The characteristics of the team design studied here included the presence of norms (e.g., Cohen, Ledford, and Spreitzer 1996), cohesion (e.g., George and Bettenhausen 1990), team size (e.g., McNamara, Luce, and Tompson 2002), and team longevity (i.e., the length of time a team's members have spent working together; Deeter-Schmelz and Ramsey 2003). Finally, team type (i.e., selling, customer service, or new-product-development teams), supplier company size, and industry were included in the model to represent environmental factors.

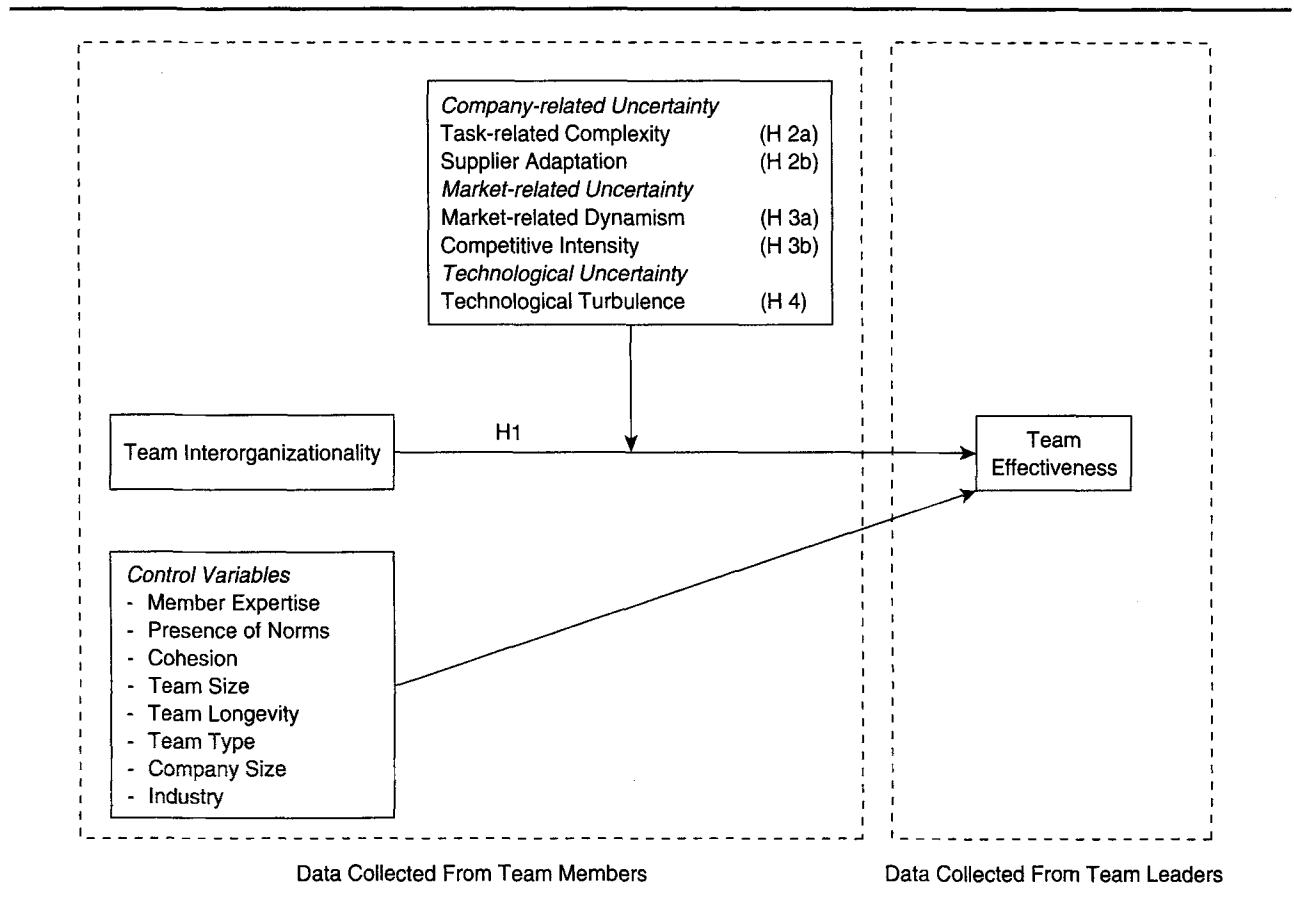
### Hypothesis Related to the Main Effect

The fundamental hypothesis of this study was related to the impact of team interorganizationality on team effectiveness. From a resource-dependence perspective, the use of interorganizational teams reflects a strategy to bridge the boundary between a supplier company and its customers. Establishing interorganizational teams enables a supplier company to get access to information and reduce the uncertainty caused by the dependence on demand from customers. Similarly, boundary theory focuses on interpersonal interactions. Members of interorganizational teams, or boundary spanners, play multiple roles at the interface between a supplier company and its business customers, related to the supplier company's image, information flows, influence on customers, and representation of the supplier to the customer (Adams 1976).

Team effectiveness will be highest when team interorganizationality is maximal, that is, when a supplier company and its customers are equally represented and power is equally distributed. If interorganizationality is low because a team is dominated by members of a supplier company, the voice of the customer likely will be ignored. If a team is dominated by members from the customer organizations, those members likely will push the team toward decisions that may run counter to the supplier company's interests and thereby reduce the team's effectiveness (from the supplier's perspective). Thus,

*Hypothesis 1: Team interorganizationality positively affects team effectiveness.*

**FIGURE 1**  
**Conceptual Framework**



Hypothesis 1 is not self-evident, in that there may be detrimental effects associated with team interorganizationality. For example, interactions between highly interorganizational teams typically are highly complex, which may reduce team effectiveness. Furthermore, channels literature indicates some evidence of a positive effect of supplier power on performance (e.g., Etgar 1976). Against this background, it could be argued that unbalanced power relationships may be more effective.

### Hypotheses Related to Moderating Effects

The choice of relevant moderator constructs also was rooted in both resource-dependence theory and boundary theory, which state that a key goal of interorganizational teams is the reduction of uncertainty for the supplier organization. The general logic pertaining to moderating effects is therefore that team interorganizationality is particularly beneficial for team effectiveness in conditions of high uncertainty (see Figure 1).

*Company-related uncertainty.* This type of uncertainty includes two moderator variables: task-related

complexity and supplier adaptation. Task-related complexity reflects the number of elements contained in a team's task and the variety of that task (Jehn 1995). It includes issues as the variety of problems that must be solved by a team and the amount of information and alternatives the team must assess to fulfill its task.

A high level of task complexity for marketing teams typically is related to highly complex customer needs. Under this circumstance, highly complex customer-related tasks have a greater need for customer-related information than do routine tasks (Jehn 1995), and this increased need to acquire and process such customer-related information makes the effectiveness-enhancing role of interorganizational teams much more important. Following this logic, the following hypothesis was formulated:

*Hypothesis 2a:* The higher the task-related complexity, the stronger the relationship between team interorganizationality and team effectiveness.

Supplier adaptation is defined as the degree to which a supplier functions according to specific demands from

individual customers (e.g., Cannon and Perreault 1999), perhaps by adapting product or service features and/or internal processes to meet those requirements. In conditions of extensive supplier adaptation, the supplier becomes dependent on the customer because of its specific investments in that relationship. Thus, control may be transferred from the supplier to the customer (Joshi and Stump 1999), which makes the supplier vulnerable to actions undertaken by the customer. Specifically, the customer can exploit the supplier's dependence in an opportunistic way (e.g., by demanding free additional services or unilateral price decreases). As a result, in the case of high supplier adaptation, the supplier company suffers an increased level of uncertainty. In such situations, the uncertainty-reducing role of interorganizational teams becomes particularly relevant. Thus,

*Hypothesis 2b:* The higher the supplier adaptation, the stronger the relationship between team interorganizationality and team effectiveness.

*Market-related uncertainty.* This type of uncertainty comprises two moderator variables: market-related dynamism and competitive intensity. Market-related dynamism is the frequency of major, market-related changes. Such changes may relate to products or services offered by competing suppliers or customer preferences. When market-related dynamism is high, there is more information to be gathered and processed by individual suppliers in the market (Homburg, Krohmer, and Workman 1999). Therefore, the uncertainty-reducing role of interorganizational teams will be more relevant in this situation than in markets with low levels of dynamism. Therefore, the following hypothesis was developed:

*Hypothesis 3a:* The higher the market-related dynamism, the stronger the relationship between team interorganizationality and team effectiveness.

Competitive intensity is the number of competitors and the amount of activities in which they engage (Jaworski and Kohli 1993). When competitive intensity is high, monitoring competitive activities becomes particularly important for suppliers. The establishment of interorganizational teams promotes information exchange between a supplier company and its customers. In addition, highly competitive markets typically make it very difficult to maintain customer loyalty. Interorganizational teams foster the development of interpersonal relationships between boundary spanners on the supplier and customer sides and therefore provide a suitable approach for maintaining customer loyalty in highly

competitive markets. Therefore, the following hypothesis was developed:

*Hypothesis 3b:* The higher the competitive intensity, the stronger the relationship between team interorganizationality and team effectiveness.

*Technological uncertainty.* This category of uncertainty was investigated through technological turbulence as the frequency of technological changes (Jaworski and Kohli 1993). A high level of technological turbulence creates significant uncertainty for a supplier in the market because technological trends must be monitored carefully. When the rate of change in the technological environment is rapid, close interactions between the supplier and customer in an interorganizational team become particularly important. Specifically, it is important to gather information about whether and how customers use new technologies. Thus,

*Hypothesis 4:* The higher the technological turbulence, the stronger the relationship between team interorganizationality and team effectiveness.

## METHOD

### Sample and Data Collection

The sample for this investigation covered several industries, as well as different types of customer-related teams in a B2B context. Specifically, the sample included the mechanical machinery, consumer goods, automotive supply, electronics, banking, and management consulting industries. The selection of these industries was based on preliminary interviews with industry experts, which revealed that customer-related teams are particularly relevant in these industries. Furthermore, the sampling procedure addressed the three different types of customer-related teams, namely, selling teams, customer service teams, and new-product-development teams.

Team members and team leaders provided the data. An important consideration in collecting the data was to obtain at least two independent team member assessments and one team leader evaluation per team.

Major companies from the various industries were randomly selected from listings maintained by industry associations. Within each of these companies, a senior manager was identified. These senior managers included those responsible for sales (and thus selling teams), research and development (new-product-development teams), and general management (customer service teams).

In the first step of the data collection, 56 managers from 56 different companies were contacted by mail and asked to participate in the study. During subsequent telephone conversations, they responded to open questions regarding their firms' participation in the study. In some

**TABLE 1**  
**Sample Characteristics**

| <i>Variable</i>                                   | <i>Number of Teams</i> |
|---------------------------------------------------|------------------------|
| Industry                                          |                        |
| Banking                                           | 35                     |
| Management consulting                             | 10                     |
| Automotive supply                                 | 39                     |
| Mechanical machinery                              | 4                      |
| Consumer goods                                    | 24                     |
| Electronics                                       | 49                     |
| Team type                                         |                        |
| Selling                                           | 57                     |
| Customer service                                  | 41                     |
| New-product development                           | 63                     |
| Number of respondents (team members/team leaders) |                        |
| 2/1                                               | 73                     |
| 3/1                                               | 45                     |
| 4/1                                               | 20                     |
| 5/1                                               | 12                     |
| > 6/1                                             | 11                     |

cases, personal visits to the companies also were made to persuade the target persons to participate.

Managers identified 5 to 10 teams in their companies and at least three individual members and the team leader from each team. As an incentive, the participating companies were offered a report that compared individual team data with industry averages, as well as a 1-day, in-house workshop addressing techniques for increasing team effectiveness (e.g., communication, conflict solving). Through this procedure, 26 companies agreed to participate.

In the second step, the team members and team leaders (identified by the managers) were contacted via an Internet-based survey. The managers identified a total of 1,429 individuals from 262 teams. In response to the initial and follow-up activities, 916 individuals from 245 teams provided data, for a response rate of 63.9 percent at the individual level and 93.5 percent at the team level. Single responses from any team were eliminated, which removed 20 additional responses, and 29 teams that produced responses from only 1 team member and 1 team leader also were eliminated. Furthermore, 35 purely intraorganizational teams (i.e., teams that consisted exclusively of members of the supplier company) were eliminated.<sup>2</sup> The final sample therefore contained 161 teams, each of which provided at least two team-member evaluations and one team-leader evaluation, for 691 total respondents, or an average number of respondents per team of 4.29. Descriptive information about the composition of the sample appears in Table 1.

### Measure Development and Assessment

The focal construct of this study, team interorganizationality, was measured according to the facets of team composition ( $IO_1$ ) and power distribution ( $IO_2$ ).

Interorganizationality with respect to team composition ( $IO_1$ ) was computed as a function of the number of team members from the supplier company ( $x_1$ ) and the customer ( $x_2$ ). Consistent with the preceding definition, team interorganizationality was computed as

$$IO_1 = \min\{x_1, x_2\} / \max\{x_1, x_2\}.$$

$IO_1$  could assume values between 0 and 1. The maximal value of 1 was assumed if  $x_1 = x_2$ , that is, if the supplier and customer sides were equally represented in a team. The minimum value of 0 was assumed if the team was composed totally of members from the supplier company (i.e.,  $x_2 = 0$ ).

The second dimension of interorganizationality ( $IO_2$ ) is the power distribution of a team. Four items were used to capture the extent to which team decisions were under the control of the supplier or the customer. For each of these decisions, assessments referred to a 7-point Likert-type scale ranging from 1 (*decision is totally under the control of team members from our company*) to 7 (*decision is totally under control of customer members in the team*). The middle point of the scale (4) represented that the *decision is equally influenced by supplier and customer company*. A complete list of items appears in Figure 2, and  $IO_2$  was computed as

$$IO_2 = 1 - 1/3 \times |x - 4|,$$

where  $x$  denotes the average assessment of the respondents across the four decision areas.

$IO_2$  was also limited to the interval from 0 to 1. The maximum value was assumed if  $x = 4$ , that is, power was evenly distributed between the two companies. The minimum value of 0 was assumed if  $x = 7$  or  $x = 1$ , indicating that the power within the team resided totally with one company.

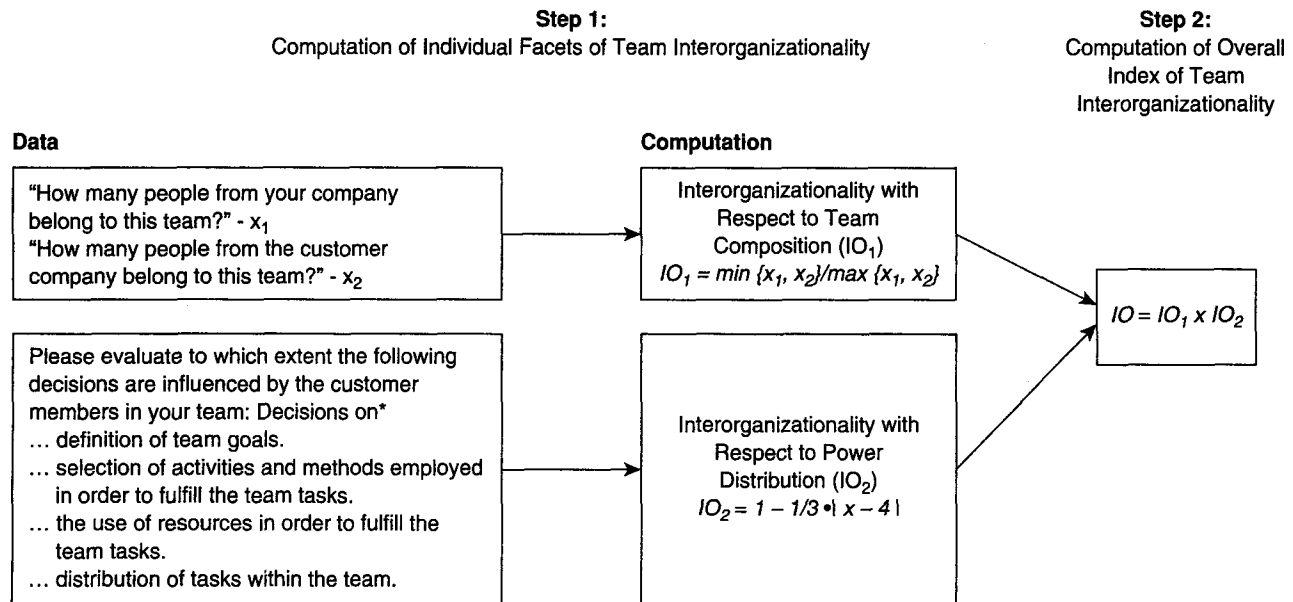
This overall index for team interorganizationality is rooted in the team theory developed by Berné (1963). Team interorganizationality assumes high values when two organizations are (nearly) equally represented and when power is equally distributed. The two facets of team interorganizationality ( $IO_1$  and  $IO_2$ ) can be combined into an overall index for interorganizationality ( $IO$ ), which is computed as

$$IO = IO_1 \times IO_2.$$

The multiplicative composition of this overall index ensures that interorganizationality is high only when both organizations are similarly represented and have comparable levels of power. Figure 2 illustrates the computation of this index.

Team effectiveness was measured by a three-item scale adapted from Campion, Papper, and Medsker (1996) and on the basis of data obtained from team

**FIGURE 2**  
**Measurement of Team Interorganizationality: Approach and Illustrative Example**



\* 7-point Likert-type scale with anchors 1 = *decision is totally under the control of team members from our company* and 7 = *decision is totally under the control of customer members in the team*. In those cases where there were no customers involved as team members, customer power was automatically rated as 1 for all of the items.

#### Illustrative Example

| Data                                                                                                                                           | Computation of Facets of Interorganizationality | Interorganizationality          |
|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------|
| Members from supplier company: $x_1 = 7$<br>Members from customer company: $x_2 = 3$                                                           | $IO_1 = .42$                                    | $IO = .42 \cdot .58$<br>$= .24$ |
| - definition of team goals: 2<br>- selection of activities and methods: 4<br>- use of resources: 3<br>- distribution of tasks: 2<br>$x = 2.75$ | $IO_2 = 1 - 1/3 \cdot  2.75 - 4 $<br>$= .58$    |                                 |

leaders (see Figure 1). Moderator and control variables were measured on the basis of scales previously developed in the literature. The main sources for these are shown in the Appendix.

To assess measurement validity, confirmatory factor analyses were run with LISREL 8 (Jöreskog and Sörbom 1993). Table 2 provides the confirmatory factor analysis results, together with additional reliability information.

Composite reliability represents the shared variance among a set of observed variables that measure an underlying construct, and a composite reliability of at least .60 is considered desirable (Fornell and Larcker 1981). This requirement was met for every factor. The values of the average variance extracted also provided satisfactory

results. As Table 2 also shows, the coefficient  $\alpha$  values were all greater than .70.

Next, the discriminant validity of the constructs was tested using the procedure suggested by Fornell and Larcker (1981). For every pair of factors, the squared correlation between them was less than the two values of the average variance extracted. A factor correlation matrix appears in Table 2.

Any research that uses multiple informants must determine whether the responses of different informants from the same team are consistent enough to be averaged for further analysis. This study assessed the acceptability of aggregating the data obtained from different members of the same team in two ways: first, an analysis of

**TABLE 2**  
**Descriptive Statistics, Reliabilities, and Intercorrelations Among Refined Measures**

|                                  | M     | SD    | CR  | AVE | Factors |       |       |       |       |       |      |      |      |       |       |       |       |       |
|----------------------------------|-------|-------|-----|-----|---------|-------|-------|-------|-------|-------|------|------|------|-------|-------|-------|-------|-------|
|                                  |       |       |     |     | 1       | 2     | 3     | 4     | 5     | 6     | 7    | 8    | 9    | 10    | 11    | 12    | 13    | 14    |
| 1. IO <sub>1</sub> (composition) | .71   | .23   |     |     | (—)     |       |       |       |       |       |      |      |      |       |       |       |       |       |
| 2. IO <sub>2</sub> (power)       | 4.73  | .65   | .89 | .68 | .13     | (.87) |       |       |       |       |      |      |      |       |       |       |       |       |
| 3. EFFECT                        | 3.08  | .71   | .89 | .51 | .24*    | .16*  | (.86) |       |       |       |      |      |      |       |       |       |       |       |
| 4. EXPERT                        | 2.97  | .65   | .96 | .59 | .01     | .34*  | .31*  | (.95) |       |       |      |      |      |       |       |       |       |       |
| 5. NORMS                         | 2.99  | .83   | .93 | .72 | .04     | .28*  | -.01  | .35*  | (.91) |       |      |      |      |       |       |       |       |       |
| 6. COHES                         | 2.82  | .66   | .90 | .65 | .02     | .43*  | .23*  | .41*  | .31*  | (.88) |      |      |      |       |       |       |       |       |
| 7. T_SIZE                        | 6.84  | 2.34  |     |     | .04     | .07   | .02   | .03   | .06   | -.01  | (—)  |      |      |       |       |       |       |       |
| 8. T_LONG                        | 14.52 | 2.99  |     |     | .01     | -.03  | .11   | .18*  | .23*  | .19*  | -.02 | (—)  |      |       |       |       |       |       |
| 9. C_SIZE                        | 11.76 | 11.84 |     |     | .12     | .01   | -.03  | .07   | -.03  | .01   | .01  | .05  | (—)  |       |       |       |       |       |
| 10. T_COMP                       | 3.06  | .83   | .83 | .53 | .12     | .22*  | .16*  | .40*  | .25*  | .26*  | .05  | .05  | .02  | (.81) |       |       |       |       |
| 11. S_ADAPT                      | 3.74  | .99   | .82 | .55 | .09     | .11   | .13   | .48*  | .12   | .01   | -.01 | -.09 | .12  | .26*  | (.80) |       |       |       |
| 12. M_DYN                        | 3.55  | 1.11  | .87 | .57 | .01     | .08   | .08   | .20*  | .11   | -.04  | -.01 | .04  | .04  | .20*  | .13   | (.84) |       |       |
| 13. COMP                         | 2.52  | .77   | .86 | .64 | .08     | .25*  | .25*  | .44*  | .22*  | .26*  | .04  | .04  | .02  | .61*  | .27*  | .23*  | (.85) |       |
| 14. T_TURB                       | 3.44  | 1.07  | .91 | .73 | .13     | .06   | .06   | .12   | .02   | .01   | -.01 | -.13 | -.10 | .19*  | .02   | .43*  | .14*  | (.87) |

NOTE: Reliabilities are in parentheses. CR = composite reliability; AVE = average variance extracted; IO<sub>1</sub> (composition) = team interorganizationality with respect to team composition; IO<sub>2</sub> (power) = interorganizationality with respect to power distribution; EFFECT = effectiveness; EXPERT = expertise; NORMS = presence of norms; COHES = cohesion; T\_SIZE = team size; T\_LONG = team longevity; C\_SIZE = company size; T\_COMP = task complexity; S\_ADAPT = supplier adaptation; M\_DYN = market dynamism; COMP = competition intensity; T\_TURB = technological turbulence. \* $p \leq .05$ .

variance tested whether the variance across teams was greater than the variance within them (e.g., George and Bettenhausen 1990). Second, the estimate of within-group interrater reliability suggested by James, Demaree, and Wolf (1981, 1993) tests for consistency within groups. The results supported the plan to average data obtained from respondents within a team.

## RESULTS

To test the hypotheses, this investigation used regression analysis. In addition to interorganizationality, the regression model contained the previously mentioned control variables as independent variables. Moderated regression analysis was used to analyze interaction effects. To minimize multicollinearity, all independent variables were mean centered.

The results appear Table 3. Hypothesis 1, which related to the effect of team interorganizationality on team effectiveness, was supported ( $\beta = .20, p < .05$ ). A second model incorporated the moderator variables (see Table 3). Hypotheses 2a, 2b, 3a, 3b, and 4 were supported: the parameter estimates associated with the interaction terms were positive and significant ( $p < .05$ ). The effect of interorganizationality on team effectiveness remained stable even when the moderating effects were incorporated into the model.

The most important control variable was team cohesiveness. Furthermore, the expertise of team members had a fairly consistent positive effect on team effectiveness. A positive effect on team effectiveness also emerged from team longevity, again consistent with findings in the

literature (e.g., Hollenbeck, Ilgen, LePine, Colquitt, and Hedlund 1998). However, the presence of norms did not significantly affect team effectiveness. One possible explanation for this finding is that the positive and negative effects of norms may compensate for each other.

## DISCUSSION

### Research-Related Implications

This study represents a first step toward a theory-based, empirically grounded understanding of team interorganizationality and provides three significant contributions to the literature. First, it offers a conceptualization and operationalization of team interorganizationality using two dimensions: team composition and power distribution. This contribution is based on theoretical work by Berné (1963), which has not been used previously in marketing research but clearly should be used more intensively. For example, this theory might prove useful for the study of cross-functional teams (e.g., Sarin and Mahajan 2001; Sethi 2000a, 2000b). In this context, the different functional backgrounds of team members could be interpreted as boundaries within the team, according to the theory developed by Berné.

Second, team interorganizationality is identified as a driver of team effectiveness on the basis of dyadic data. Therefore, further research on team success factors should include team interorganizationality as a driver of team performance. Researchers following this recommendation also can use the measurement instrument developed herein for the construct of team interorganizationality.

**TABLE 3**  
**Results of Regression Analysis**

| <i>Independent Variable (team member data)</i>          | <i>Team Effectiveness as Dependent Variable (team leader data)</i> |                               |
|---------------------------------------------------------|--------------------------------------------------------------------|-------------------------------|
|                                                         | <i>Without Moderator Effects</i>                                   | <i>With Moderator Effects</i> |
| Main effect                                             |                                                                    |                               |
| Interorganizationality                                  | .19*                                                               | .17*                          |
| Direct effects of moderator variables                   |                                                                    |                               |
| Task-related complexity                                 |                                                                    | .05                           |
| Supplier adaptation                                     |                                                                    | .26*                          |
| Market-related dynamism                                 |                                                                    | .01                           |
| Competitive intensity                                   |                                                                    | .03                           |
| Technological turbulence                                |                                                                    | .05                           |
| Interaction effects                                     |                                                                    |                               |
| Interorganizationality × Task-Related Complexity        |                                                                    | .22*                          |
| Interorganizationality × Supplier Adaptation            |                                                                    | .15*                          |
| Interorganizationality × Market-Related Dynamism        |                                                                    | .16*                          |
| Interorganizationality × Competitive Intensity          |                                                                    | .19*                          |
| Interorganizationality × Technological Turbulence       |                                                                    | .11*                          |
| Control variables                                       |                                                                    |                               |
| Expertise of members                                    | .18*                                                               | .17*                          |
| Presence of norms                                       | -.05                                                               | .04                           |
| Cohesion                                                | .19*                                                               | .20*                          |
| Team size                                               | -.02                                                               | .01                           |
| Team longevity                                          | .14*                                                               | .15*                          |
| Company size                                            | .01                                                                | .07                           |
| Team type <sup>a</sup>                                  |                                                                    |                               |
| Customer selling                                        | .05                                                                | .08                           |
| New-product development                                 | .03                                                                | .04                           |
| Industry <sup>b</sup>                                   |                                                                    |                               |
| Banking                                                 | .05                                                                | .09                           |
| Management consulting                                   | .06                                                                | .04                           |
| Automotive supply                                       | .03                                                                | .08                           |
| Mechanical machinery                                    | .01                                                                | -.03                          |
| <i>F</i>                                                | 4.03*                                                              | 3.99*                         |
| <i>R</i> <sup>2</sup> (adjusted <i>R</i> <sup>2</sup> ) | .23 (.15)                                                          | .33 (.28)                     |

NOTE: Standardized coefficients are reported.

a. Customer service teams as reference category.

b. Consumer goods industry as reference category.

\**p* ≤ .05.

A third important implication of this study is to show that there are contingency factors that systematically strengthen the relationship between team interorganizationality and team effectiveness. These include aspects of company-related, market-related, and technological uncertainty. The findings regarding these moderators are consistent across different types of teams, so they obviously should be considered in research on the success factors of marketing teams.

### Managerial Implications

This study's most important finding for managers is that team interorganizationality positively affects a team's effectiveness. Thus, managers should use various types of interorganizational teams for marketing tasks, such as customer relationship management and new product

development. Specifically, managers should involve customers in such teams.

The findings also encourage managers to shift power within these teams to customers. For a given team structure, team interorganizationality is highest when the power is equally distributed, and though most managers tend to dislike giving away power to actors outside the organization (e.g., customers), this study clearly shows that the use of truly interorganizational teams (which involve power balance) positively affects team effectiveness from the supplier's perspective.

This study furthermore informs managers about the conditions in which high team interorganizationality is especially beneficial as a driver of team effectiveness. Specifically, a high level of team interorganizationality is fruitful when the task to be achieved is highly complex, as might be the case for management consulting firms. In

addition, managers should build highly interorganizational teams to manage those customer relationships for which they have undertaken significant adaptations. This finding may be particularly relevant for industries such as the automotive supply industry, in which companies often make important adaptations to support their collaboration with specific customers.

The findings suggest that interorganizational teams are particularly beneficial when uncertainty is high (which may be caused by market-related dynamism, competitive intensity, or technological turbulence). Therefore, managers should understand the use of interorganizational teams as a means to cope with external uncertainty.

### Limitations and Avenues for Further Research

This study investigated the direct performance impact of team interorganizationality. Additional research might investigate the extent to which team interorganizationality moderates the link between team characteristics and team performance. The interesting question in this context is whether the performance of highly interorganizational teams is driven by mechanisms other than the performance of intraorganizational teams.

This study was based on cross-sectional data collected at one point in time, but it would be interesting to study interorganizational teams from a dynamic perspective to determine whether the formation of interorganizational teams differs from the formation of intraorganizational teams in terms of duration, for example.

Moreover, this analysis was limited to data obtained from the supplier company. Additional research should analyze the phenomenon of interorganizational teams using data from both sides of the dyad. Research of this type might for example investigate the impact of team interorganizationality on customer perceptions of relationship characteristics, such as satisfaction, trust, and commitment, or the percentage of business given to the supplier. In summary, though this study introduces the new phenomenon of team interorganizationality to the marketing discipline, it also identifies several future research directions for studying this construct.

## APPENDIX

### Measures and Items

#### *Team Performance*

Team effectiveness<sup>a</sup> (expanded scale from Campion et al. 1996)

The team . . .

- . . . mostly achieves the team goals.
- . . . performs very well.
- . . . overall is very successful.

#### *Control variables*

Expertise of members<sup>a</sup> (expanded scale from Doney and Cannon 1997)

Team members . . .

- . . . know their company's product/service range very well.
- . . . are typically very well informed.
- . . . are very knowledgeable.
- . . . know about the newest developments (new products, new technologies etc.).
- . . . know the competitors' product/services range very well.
- . . . are typically able to understand complex situations.
- . . . work out solutions which are very well thought through.
- . . . know how to use problem solving techniques.
- . . . have a high level of conceptual abilities.
- . . . have organized their work processes very well.
- . . . have a well structured time management.
- . . . have skills in project management.

Presence of norms<sup>a</sup> (adapted from Denison, Hart, and Kahn 1996)

In our team . . .

- . . . it is clear what is acceptable behavior and what is not acceptable.
- . . . behavior is very orderly—it is clear what members are expected to do and they do it.
- . . . we have clear standards for the behaviors of the team members.
- . . . members know very well which behavior is expected from them.
- . . . we have standards which regulate behavior among group members.

Cohesion<sup>a</sup> (adapted from Jehn 1995)

In our team . . .

- . . . members generally like the other members.
- . . . members like to stay in this team.
- . . . there is a high group spirit.
- . . . members appreciate each other personally.
- . . . members enjoy being a member of this work unit.

#### *Team size*

How many members does your team actually have? \_\_\_\_ members.

Team longevity (adapted from Sethi 2000a)

For how many months does your team work together? \_\_\_\_ months.

#### *Company size*

About how many employees does your company actually have? \_\_\_\_ employees.

#### *Team type*

Objective measure (data obtained from the supplier company).

#### *Moderator variables*

Task-related complexity<sup>a</sup> (adapted from Jehn 1995)

The tasks of our team . . .

- . . . include much variety.
- . . . mainly consist of solving complex problems.
- . . . hardly consist of routine work.

- ... require the assessment of a large amount of information/alternatives.
- ... consist of numerous different elements.

Supplier adaptation<sup>a</sup> (adapted from Hallén, Johansson, and Seyed-Mohamed 1991)

The procedures of our company are highly customized.

The products/services are strongly adapted to our customers' needs.

We can typically not market to other customers the products/services sold to a customer.

We produce the products/services exactly according to our customers' specifications.

Market-related dynamism<sup>a</sup> (adapted from Homburg et al. 1999)

In our industry . . .

- ... products/services of your competitors change frequently.
- ... sales strategies by your competitors change frequently.
- ... customer preferences in product features change frequently.
- ... customer preferences in price/performance features change frequently.
- ... competitors change frequently.

Competitive intensity<sup>a</sup> (adapted from Jaworski and Kohli 1993)

In our market . . .

- ... competition is cutthroat.
- ... price competition is hallmark.
- ... one hears of a new competitive move almost every day.
- ... competitors fight each other to win big customers.

Technological turbulence<sup>a</sup> (Jaworski and Kohli 1993)

In our industry . . .

- ... the technology is changing rapidly.
- ... technological changes provide big opportunities in our industry.
- ... a large number of new product ideas have been made possible through technological breakthroughs in our industry.

a. 7-point Likert-type scale ranging from 1 (*strongly agree*) to 7 (*strongly disagree*) were used.

## NOTES

1. Interorganizational teams differ from purely intraorganizational teams in several important aspects. First, whereas purely intraorganizational teams typically focus on the goals of one company, interorganizational teams must consider the goals of two companies (e.g., Halal 1994; Nohria and Berkley 1994). Because these two goal structures usually are not totally consistent, goal conflict is inherent in interorganizational teams, which is not the case for purely intraorganizational teams. Second, members of interorganizational teams are embedded in two different organizational cultures, so they likely are used to different interaction styles in their respective work environments. Thus, interaction is typically more difficult and complex in interorganizational teams than in purely intraorganizational teams. Therefore, it is problematic to generalize knowledge about intraorganizational teams to interorganizational teams.

2. These eliminations followed the suggestion of an anonymous reviewer and the editor.

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## ABOUT THE AUTHOR

**Ruth Maria Stock** (ruth.stock@uni-hohenheim.de) is a professor of business administration and management at the University of Stuttgart-Hohenheim, Germany. She holds a master's degree in psychology from the University of Hagen, Germany, a PhD in business administration from the University of Mannheim, Germany, and a habilitation degree from the University of Bundeswehr, Hamburg, Germany. She has published in various forums, including the *Journal of the Academy of Marketing Science*, *Psychology & Marketing*, and *Journal of Business-to-Business Marketing*. Her main research areas include market-oriented management and business-to-business marketing.